IGCSE · Cambridge (CIE) · Maths

2 hours 21 questions

Non-Calculator Questions

Tree Diagrams & Multiple Events

Probability Tree Diagrams / Combined Probabilities / Conditional Probability / Combined Conditional Probabilities

Medium (5 questions)	/18
Hard (8 questions)	/50
Very Hard (8 questions)	/54
Total Marks	**/122**

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Medium Questions

1 $\boxed{1}$ $\boxed{2}$ $\boxed{3}$ $\boxed{4}$ $\boxed{5}$

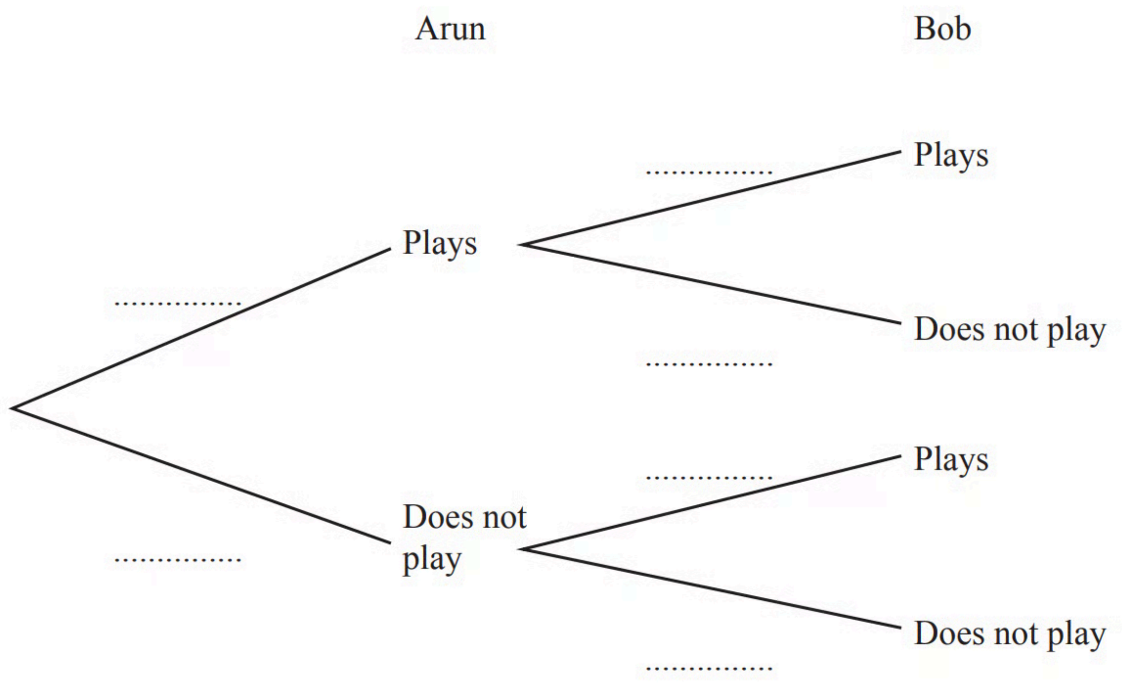
The diagram shows five cards.
Two of the cards are taken at random, without replacement.

Find the probability that both cards show an even number.

(2 marks)

2 (a) On any Saturday, the probability that Arun plays football is $\frac{3}{4}$. On any Saturday, the probability that Bob plays football is $\frac{2}{5}$.

Complete the tree diagram.



(2 marks)

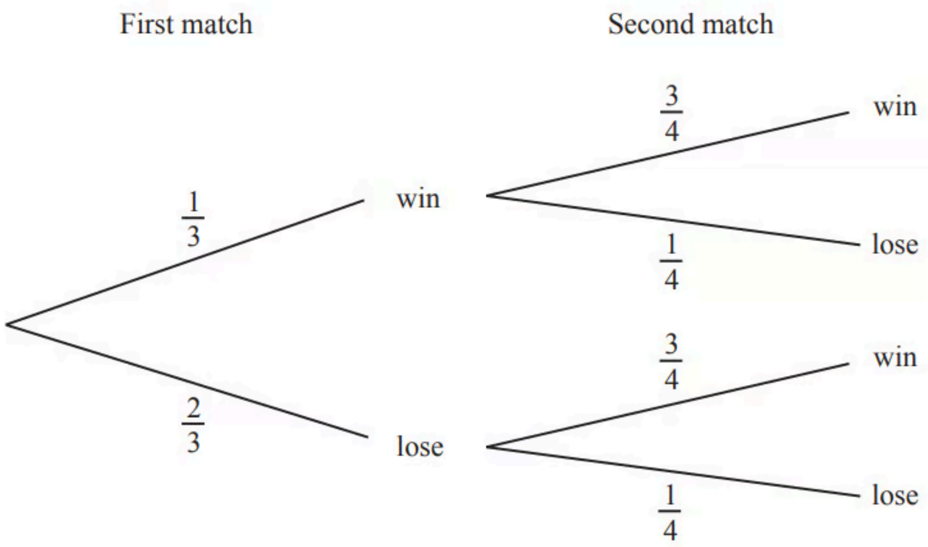
(b) Calculate the probability that, one Saturday, Arun and Bob both play football.

(2 marks)

(c) Calculate the probability that, one Saturday, either Arun plays football or Bob plays football, but not both.

(3 marks)

3 A soccer team plays two matches.
The tree diagram shows the probability of the team winning or losing the matches.



Find the probability that the soccer team wins at least one of the two matches.

(3 marks)

4

P	O	S	S	I	B	I	L	I	T	Y

Morgan picks two of these letters, at random, **without** replacement.
Find the probability that he picks

i) the letter Y first,

[1]

ii) the letter B then the letter Y.

[2]

(3 marks)

5 A bag contains 4 red marbles and 2 yellow marbles.
Behnaz picks two marbles at random without replacement.

Find the probability that

i) the marbles are both red,

[2]

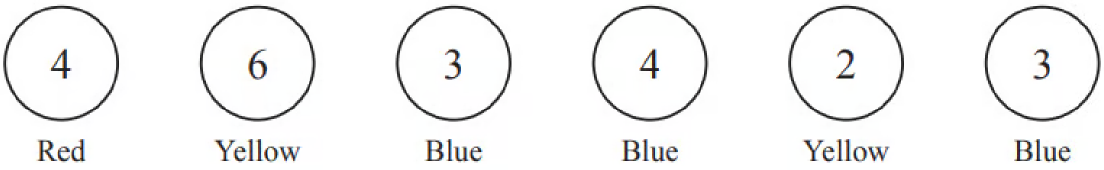
ii) the marbles are not both red.

[1]

(3 marks)

Hard Questions

1



The diagram shows six discs. Each disc has a colour and a number.

Two of the six discs are picked at random **without** replacement. Find the probability that

i) both discs have the number 3,

[2]

ii) both discs have the same colour.

[3]

(5 marks)

2

Morgan picks two of these letters, at random, **without** replacement.
Find the probability that he picks two letters that are the same.

(3 marks)

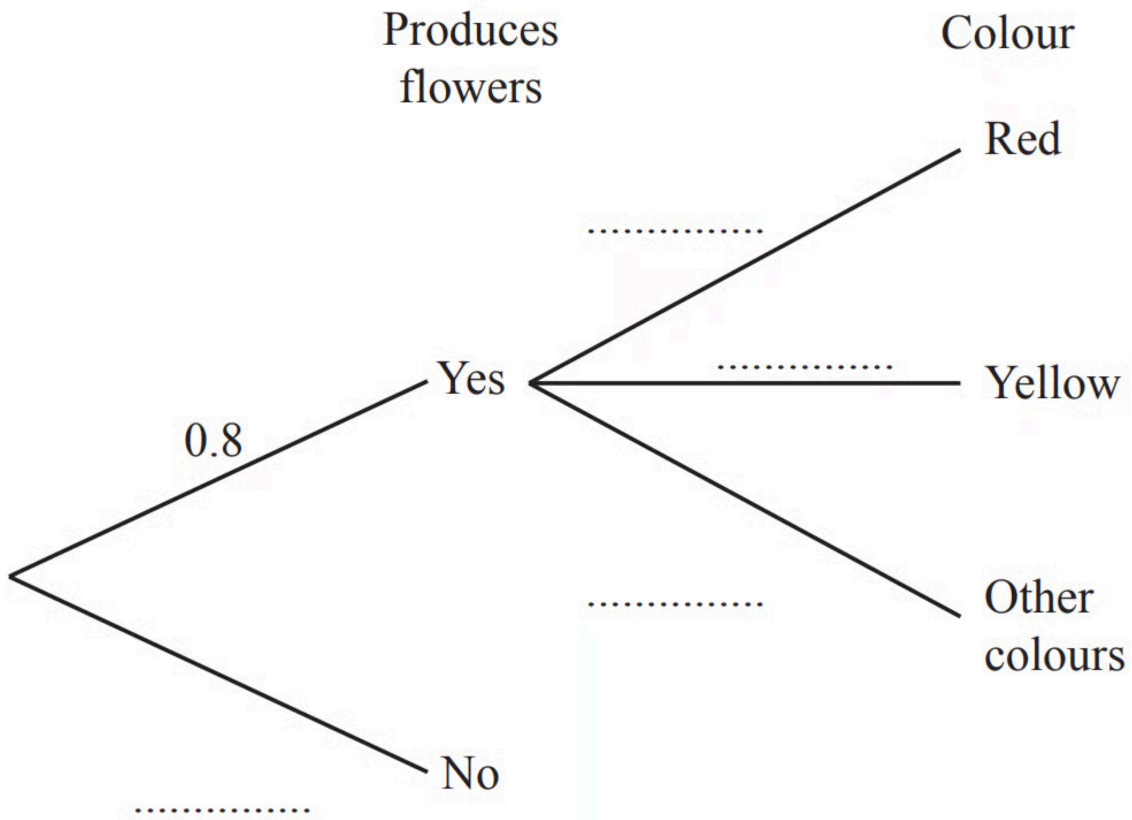
3 A bag contains 5 blue marbles and 2 green marbles.
Bryn picks one marble at random without replacement.
If this marble is not green, he picks another marble at random without replacement.
He continues until he picks a green marble.

Find the probability that he picks a green marble on his first, second or third attempt.

(4 marks)

4 (a) Tanya plants some seeds.
The probability that a seed will produce flowers is 0.8.
When a seed produces flowers, the probability that the flowers are red is 0.6 and the probability that the flowers are yellow is 0.3.

Complete the tree diagram.



(2 marks)

(b) Find the probability that a seed chosen at random produces red flowers.

(2 marks)

(c) Tanya chooses a seed at random.
Find the probability that this seed does not produce red flowers and does not produce yellow flowers.

(3 marks)

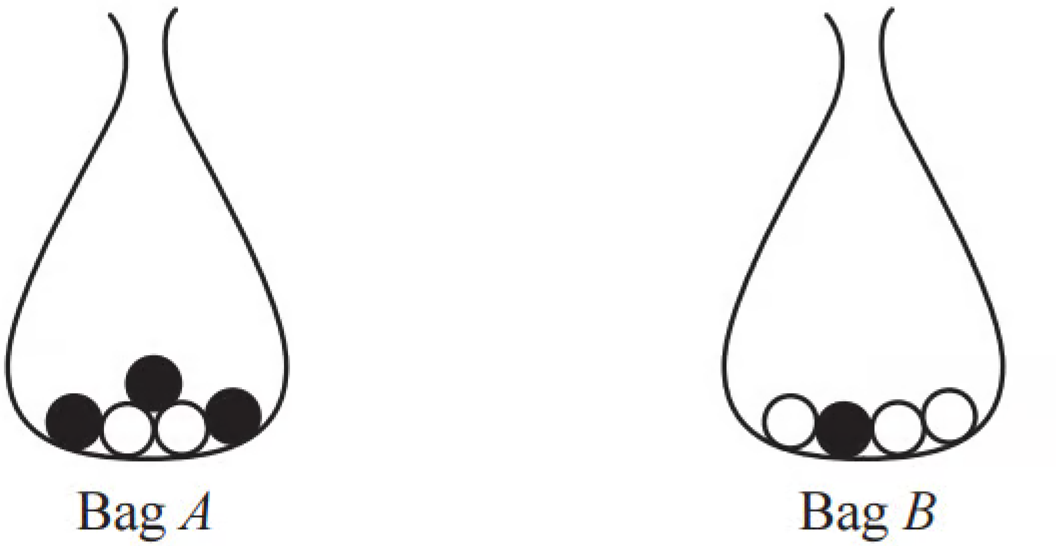
(d) Two of the seeds are chosen at random.
Find the probability that one produces flowers and one does not produce flowers.

(3 marks)

5 Esme has a bag with 5 green counters and 4 red counters.
She takes three counters at random from the bag without replacement.
Work out the probability that the three counters are all the same colour.

(4 marks)

6 (a)



Bag A contains 3 black balls and 2 white balls.
Bag B contains 1 black ball and 3 white balls.
A ball is taken at random from each bag.

i) Show that a black ball is more likely to be taken from bag *A* than from bag *B*.

[1]

ii) Find the probability that the two balls have different colours.

[3]

(4 marks)

(b) The balls are returned to their original bags.
Three balls are taken at random from bag *A*, without replacement.

i) Find the probability that they are all black.

[2]

ii) Find the probability that they are all white.

[1]

(3 marks)

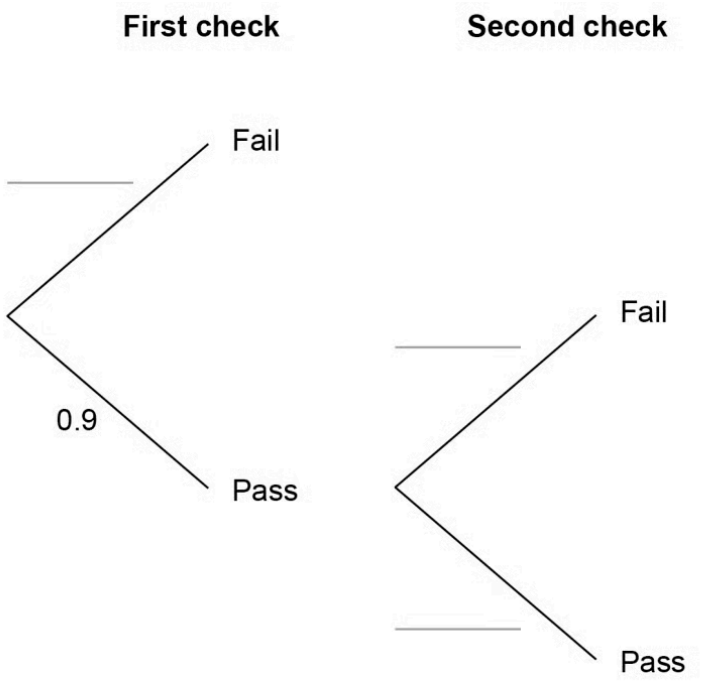
(c) The balls are returned to their original bags.
A ball is taken at random from bag *A* and its colour is recorded. This ball is then placed in bag *B*.
A ball is then taken at random from bag *B*.
Find the probability that the ball taken from bag *B* has a different colour to the ball taken from bag *A*.

(3 marks)

7 (a) Items made at a factory have to pass two checks.

90% pass the first check.
The items that fail are scrapped. 99% of the items that pass the first check pass the second check.
The items that fail are scrapped.

Complete the tree diagram.



(2 marks)

(b) An item is chosen at random before the checks.

Work out the probability that the item is scrapped.

(3 marks)

8 (a) A bag contains 3 red, 4 blue, and 2 green counters.

A counter is taken and not replaced. Another counter is then taken.

Work out the probability that both counters are blue.

(3 marks)

(b) Work out the probability that the two counters taken are of different colours.

(6 marks)

Very Hard Questions

1

P	O	S	S	I	B	I	L	I	T	Y

Morgan picks three of these letters, at random, **without** replacement.
Find the probability that

i) all three letters are the same,

[2]

ii) exactly two of the three letters are the same,

[5]

iii) all three letters are different.

[2]

(9 marks)

2 (a) Harris is taking a driving test.
The probability that he passes the driving test at the first attempt is 0.6.
If he fails, the probability that he passes at any further attempt is 0.75.

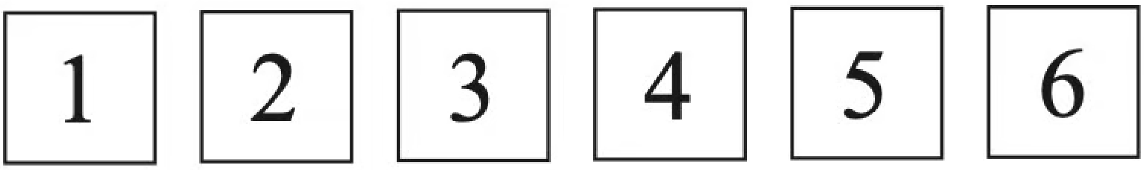
Calculate the probability that Harris passes the driving test at the second attempt.

(2 marks)

(b) Calculate the probability that Harris takes no more than three attempts to pass the driving test.

(2 marks)

3 Suleika has six cards numbered 1 to 6.



Suleika takes two cards at random, without replacement.

i) Find the probability that the sum of the numbers on the two cards is 5.

[3]

ii) Find the probability that at least one of the numbers on the cards is a square number.

[3]

(6 marks)

4 (a) The probability that Andrei cycles to school is r.

Write down, in terms of r, the probability that Andrei **does not** cycle to school.

(1 mark)

(b) The probability that Benoit **does not** cycle to school is $1.3 - r$. The probability that both Andrei and Benoit **do not** cycle to school is 0.4.

i) Complete the equation in terms of r.

$$(.........................) \times (.........................) = 0.4$$

[1]

ii) Show that this equation simplifies to $10r^2 - 23r + 9 = 0$.

[3]

iii) Solve by factorisation $10r^2 - 23r + 9 = 0$.

r = ... or r = ... [3]

iv) Find the probability that Benoit **does not** cycle to school.

[1]

(8 marks)

5

The diagram shows 5 cards.

Donald chooses two of the five cards at random, without replacement.
He works out the total number of dots on these two cards.

i) Find the probability that the total number of dots is 5.

[3]

ii) Find the probability that the total number of dots is an odd number.

[3]

(6 marks)

6 (a) Angelo has a bag containing 3 white counters and x black counters.
He takes two counters at random from the bag, without replacement.
Complete the following statement.

The probability that Angelo takes two black counters is $\dfrac{x}{x+3} \times \dfrac{\dots\dots\dots\dots}{\dots\dots\dots\dots}$

(2 marks)

(b) The probability that Angelo takes two black counters is $\frac{7}{15}$.

i) Show that $4x^2 - 25x - 21 = 0$.

[4]

ii) Solve by factorisation $4x^2 - 25x - 21 = 0$.

x = ... or x = ... [3]

iii) Write down the number of black counters in the bag.

[1]

(8 marks)

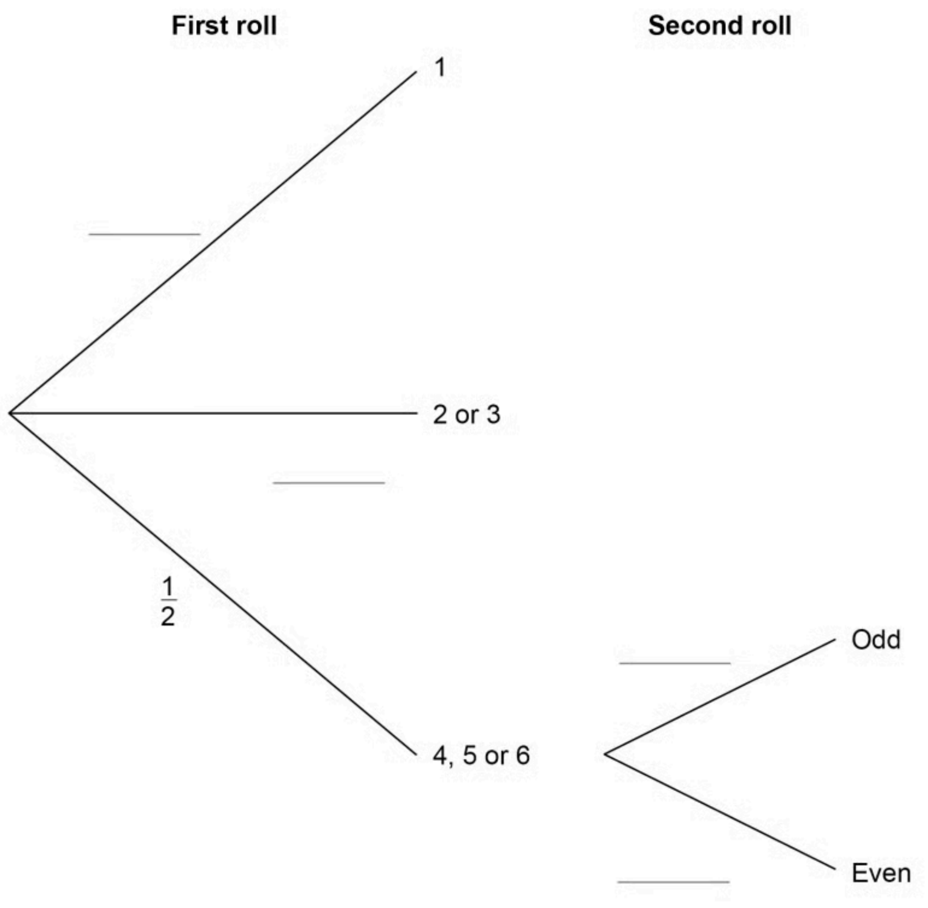
7 (a) Anna plays a game with an ordinary, fair dice.

If she rolls 1 she wins. If she rolls 2 or 3 she loses. If she rolls 4, 5 or 6 she rolls again.

When she has to roll again,

if she rolls an odd number she wins if she rolls an even number she loses.

Complete the tree diagram with the four missing probabilities.



(2 marks)

(b) Is Anna more likely to win or to lose?

You **must** work out the probability that she wins.

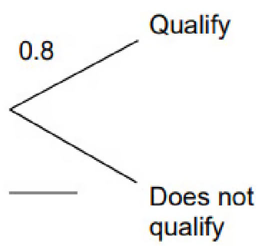
(4 marks)

8 (a) On Friday, Greg takes part in a long jump competition. He has to jump at least 7.5 metres to qualify for the final on Saturday.

- He has up to three jumps to qualify.
- If he jumps at least 7.5 metres he does **not** jump again on Friday.

Each time Greg jumps, the probability he jumps at least 7.5 metres is 0.8 Assume each jump is independent. Complete the tree diagram.

First jump	Second jump	Third jump



(2 marks)

(b) Work out the probability that he does **not** need the third jump to qualify.

(2 marks)